

Early intervention for auto lending, on one governed
lakehouse | CRO sponsor + Collections & Risk Ops
owner

The business outcome

Predict which current loans roll 30+ days past due NEXT cycle, and give collections a ranked queue to intervene before the roll.

Result the business feels: lower net charge-offs and a more efficient collections team.

No change to how loans are booked.

The problem, in your terms

Collections is reactive: a loan is worked only after it is already delinquent, when cure is harder.

Capacity is finite and spread without prioritization.

Cost = early-stage roll rate, roll-to-charge-off, net charge-offs.

On this 5,000-loan book (~\$134.6M), 3.2% of current loans roll to 30+ DPD each cycle.

One integrated journey, not six silos

Lakeflow ingests raw loan + payment data (bronze to silver with data-quality rules).

Unity Catalog governs it (comments, governed tag policy, grants, lineage).

ML + GenAI score every current loan and write a plain-language briefing.

Lakebase serves the scored queue for low-latency access.

Genie answers questions in plain English.

A Databricks App surfaces the queue, briefing, and case actions.

Proof it works (from the actual build)

Next-month 30+ DPD model, registered in Unity Catalog.

ROC AUC 0.74; risk ranks monotonically with FICO (0.5% to 11.2%).

3.7x lift in the top 5%: the highest-risk slice is ~3.7x more likely to actually roll.

Today's queue: 153 High-risk loans, \$4.4M exposure, with an AI briefing + recommended action.

Quantified impact (your KPIs)

Collections hit rate: 11.8% vs 3.2% baseline — ~3.7x more productive contacts.

Intervene on the 153 High-risk loans each cycle instead of waiting.

Illustrative: cure 25% of the \$4.4M High-risk exposure at ~55% loss-given-default
= ~\$0.6M charge-offs avoided per cycle (validate assumptions).

No new cost to book loans.

Value for the executive sponsor (CRO)

Lower net charge-offs and provisioning by shifting collections from reactive to predictive.

One governed platform from raw data to app: auditable lineage, governed tag policies, access control.

Scales to pricing, line management, and recovery on infrastructure you already run.

Value for the collections & risk ops lead

A ranked daily queue instead of a flat list — work the riskiest dollars first.

An AI briefing that explains the queue and recommends an action.

Self-serve answers via Genie, no SQL.

Case workflow built in: open and assign a case from the same screen.

Why Databricks — why integrated matters

The hand-offs between ingest, governance, ML, serving, NL, and app are the usual failure points.

Here they are one platform, one security model, one copy of governed data.

From a raw file to a business decision with no data movement to bolt-on tools.

Next steps

Validate the lift on your real (scrubbed) charge-off history.

Agree the intervention playbook for the High tier.

Pilot with collections for one cycle; measure roll-rate and hit-rate deltas.

Expand the governed lakehouse pattern to the next risk use case.